

1. EXECUTIVE SUMMARY & CORE HYPOTHESIS

Hypothesis Investigated: "After a significant one-day fall in NIFTY, the market tends to recover over the next few trading days."

Key Quantitative Findings:

- 1. Statistical Insignificance: After a drop $\leq -2.0\%$ (N=200), average 5-day forward return is $+0.32\%$ vs an unconditional baseline of $+0.21\%$ (Abnormal Return = $+0.10\%$, Welch $t = +0.25$, $p = 0.8031$; 10,000-sample Block Bootstrap $p = 0.7863$). The bounce is statistically indistinguishable from random market drift.
- 2. Survival Hazard Rate: By Day 1, probability of 100% recovery is only 14.5%. By Day 5, full recovery probability is 49.0%. Over 51% of all crash days remain unrecovered after an entire trading week.
- 3. Execution Trap: Mean overnight gap is $+0.15\%$ (58% positive). When entering at Day T+1 Open, 5-day return drops to $+0.16\%$ (Abnormal Return = -0.06% , $p = 0.8836$). The edge is locked in the overnight gap and cannot be captured during market hours.
- 4. Severe Regime Asymmetry: In bull regimes (Price > 200 SMA), 3-day recovery is $+0.65\%$ (59.3% win rate). In bear regimes (Price < 200 SMA, 73% of all crashes), recovery collapses to $+0.14\%$ (52.7% win rate) due to downside momentum and volatility clustering.
- 5. Economic Non-Viability: Net of 0.07% Indian statutory costs (STT, NSE fees, 4 bps slippage), blind 5-day holding loses -7.16% with a -46.18% maximum drawdown.

2. EXPERIMENTAL DESIGN & DEFINITIONS

- * Event Definition: Daily Close-to-Close Return $\leq -2.0\%$ (N = 200).
- * Baseline Model: Unconditional rolling H-day returns across all 4,664 days.
- * Dual Execution Models: Model A (Day T Close MOC) vs Model B (Day T+1 Open 9:15 AM).
- * Out-of-Sample Split: In-Sample 2007-2018 (N=149) vs Out-of-Sample 2019-2026 (N=51).

3. EMPIRICAL RESULTS & HYPOTHESIS TESTING

Table 1: Benchmark Event Returns vs Unconditional Baseline (N = 200)

Horizon	Baseline Mean	Model A Mean	Abnormal	Welch t (p-val)	Boot p	Sig?
1d	+0.04%	+0.11%	+0.06%	+0.35 (0.7291)	0.4804	Fail
2d	+0.09%	+0.09%	+0.00%	+0.01 (0.9946)	0.9925	Fail
3d	+0.13%	+0.28%	+0.15%	+0.47 (0.6405)	0.5426	Fail
5d	+0.21%	+0.32%	+0.10%	+0.25 (0.8031)	0.7863	Fail
10d	+0.42%	+0.60%	+0.18%	+0.34 (0.7364)	0.7546	Fail

Table 2: Survival Analysis (Hazard Rate of Recovery by Day K)

Elapsed Days (K)	Cumulative Prob of 50% Recovery	Cumulative Prob of 100% Recovery
Day 1	43.5%	14.5%
Day 2	59.0%	26.0%
Day 3	65.5%	39.0%
Day 5	72.0%	49.0%
Day 10	81.5%	65.5%
Day 20	85.5%	74.5%

4. CRITICAL FALSIFICATION & RISK ANALYSIS

- 1. The 200 SMA Bear Market Trap:
73% of large drops occur below the 200-day moving average. In bear markets, one-day falls exhibit severe downside momentum and volatility clustering rather than mean reversion. Dip-buying during bear regimes causes catastrophic compounding losses.
- 2. Microstructure Findings (Capitulation vs Absorption):
 - * Volume Climax alone ($> 1.5 \times 20d$ mean): 5-day return is -1.34% (N=15). Volume surges reflect institutional liquidation rather than exhaustion.
 - * Pin-bar Lower Shadow Absorption: 5-day return is +1.57% (N=25). Rebound occurs only when buyers step in before the close to defend intraday lows.
- 3. Backtest Simulation (1,000,000 INR Capital, Net of 0.07% Indian Friction):
 - * Model A (Close Entry, Blind 5d Exit): Net Return: -7.16% | Max Drawdown: -46.18%
 - * Model B (Open Entry, Blind 5d Exit): Net Return: -16.71% | Max Drawdown: -45.75%
 - * Risk-Managed ($1.5 \times ATR$ Stop + TP): Net Return: +0.82% | Win Rate: 54.4%

5. FINAL CONCLUSION & QUANT RECOMMENDATION

- The empirical evidence FIRMLY REJECTS the unconditioned hypothesis:
- 1. Post-drop forward returns are statistically indistinguishable from baseline market drift across all tested horizons ($p > 0.60$).
 - 2. Survival analysis shows over 51% of crashes remain unrecovered after 5 trading days.
 - 3. The marginal edge is trapped in overnight gaps and destroyed by bear market momentum.
 - 4. After standard Indian statutory costs, unconditioned mean reversion is economically unviable.

Quantitative Recommendation:
An actionable trading model must condition entries on lower shadow price absorption, restrict trades strictly to bull regimes ($Price > 200 SMA$), and enforce $1.5 \times ATR$ trailing stops.